

ייבוא ספריית turtle

על מנת שנוכל לעבוד עם הצייר

```
import turtle
```

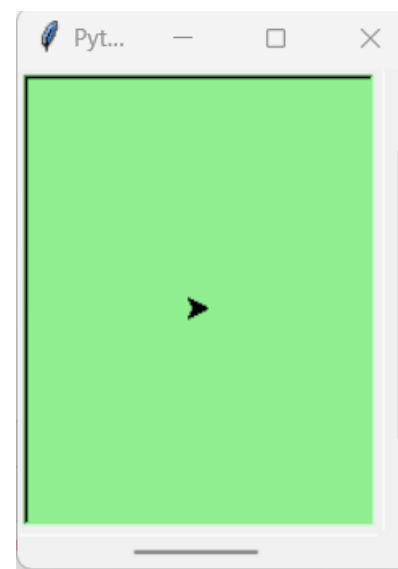
הגדרות מסך

```
# על מנת שנוכל להשתמש בספרייה
import turtle

#אתחול
wn = turtle.Screen()

# גודל המסך
#(ציקן x, ציר y)
wn.setup(200,250)

#צבע הרקע של הדמות
# אפשר להגדיר כל צבע
# ברירת המחדל היא צבע לבן
wn.bgcolor('LightGreen')
```



הגדרות הדמות

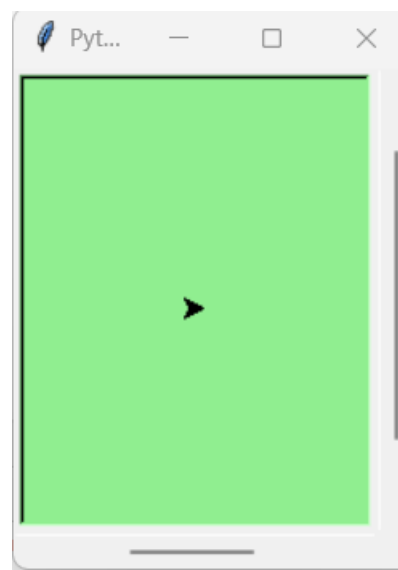
הדמות שזזה ומציירת על המסך

שם הדמות שלנו

```
player = turtle.Turtle()
```

על מנת שהמסך לא יעלם

```
turtle.done()
```



צורת הדמות

הצורה של הדמות#

אפשרויות: #

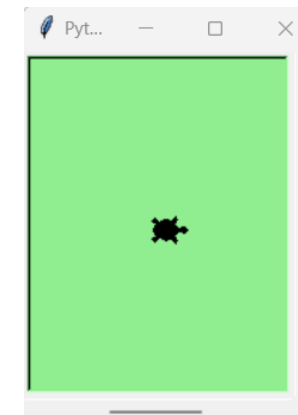
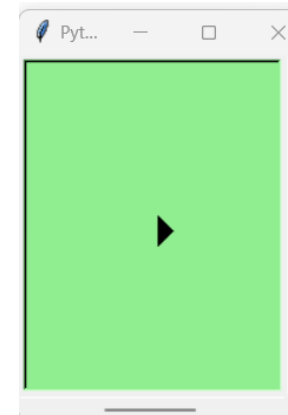
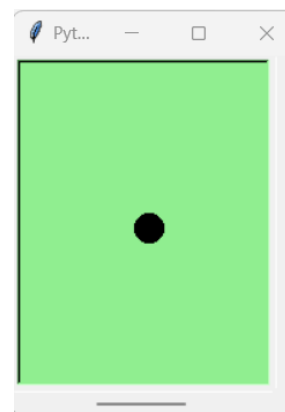
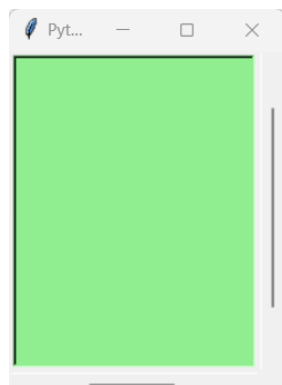
'turtle', 'arrow', 'circle', 'triangle', 'classic', 'blank'

classic המחדל היא

אפשר גם blank - ללא צורה #

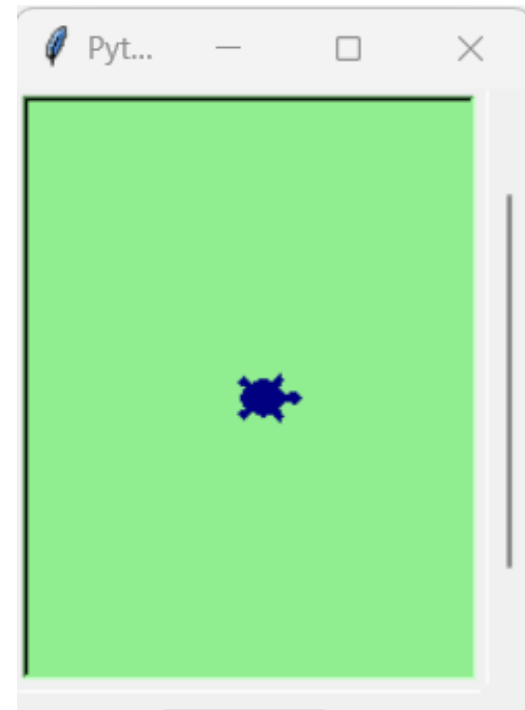
player.shape('blank')

turtle.done()



צבע הדמות

```
player.color( 'Navy' )
```



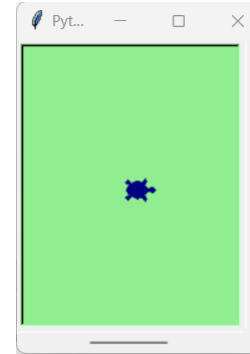
עובי הדמות

עובי הדמות

מתחיל מ 1

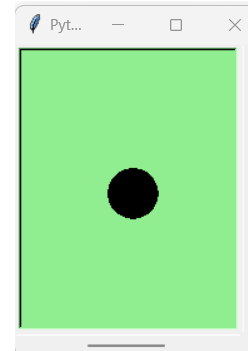
```
player.shape('turtle')
```

```
player.shapesize(1)
```



```
turtle.shape("circle")
```

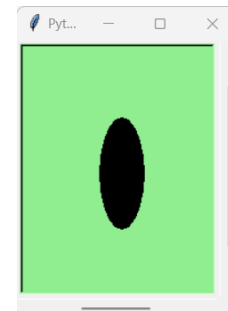
```
turtle.shapesize(2)
```



אפשר גם למתוח

```
turtle.shape("circle")
```

```
turtle.shapesize(5,2)
```



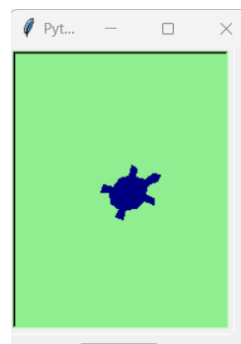
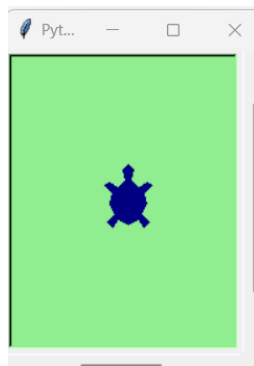
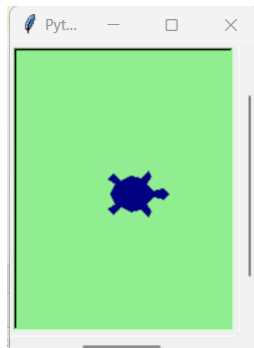
פנייה שמאלה\ימינה

```
# פניה ימינה\שמאלה
```

```
# מעלות 0-360. הדמות סובבת סביב הציר של עצמה
```

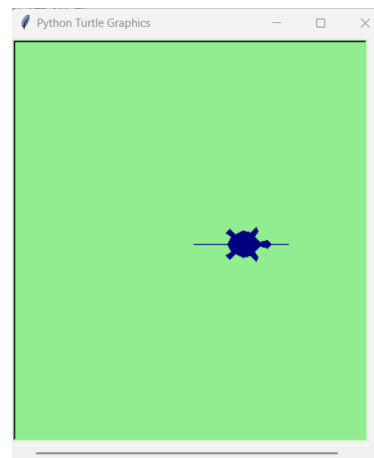
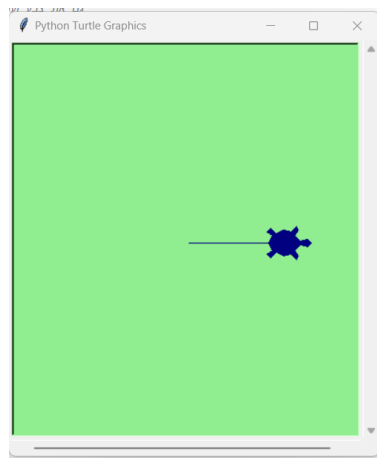
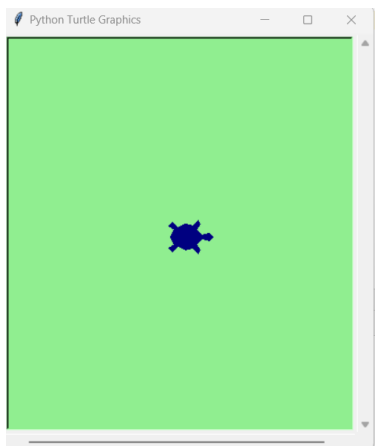
```
player.left(90)
```

```
player.right(60)
```



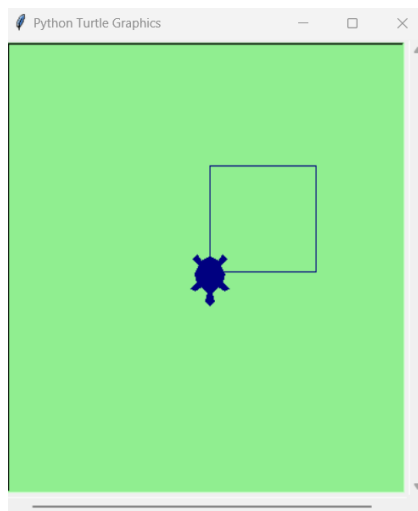
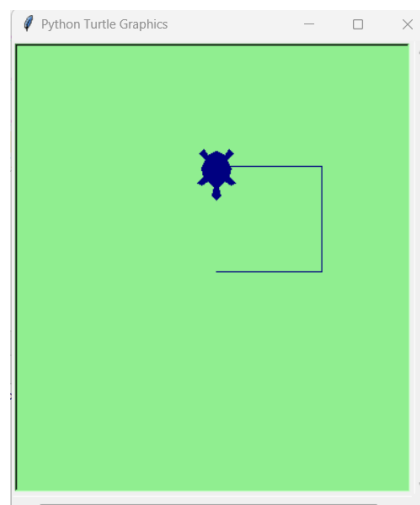
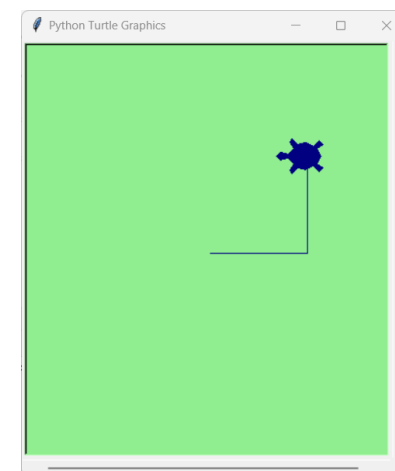
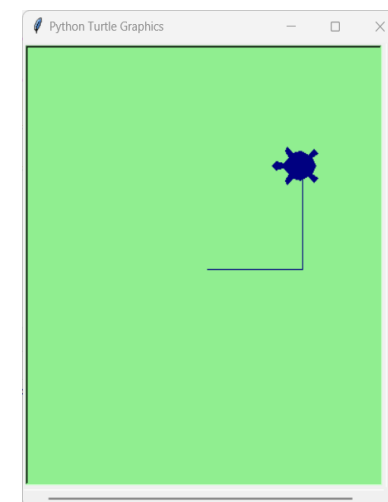
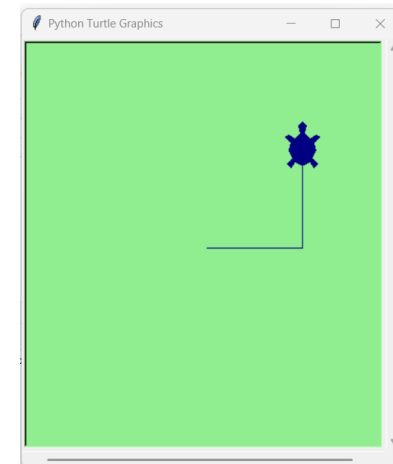
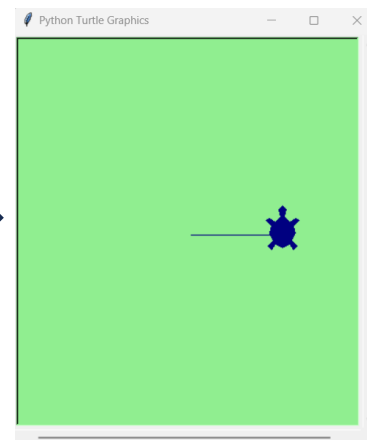
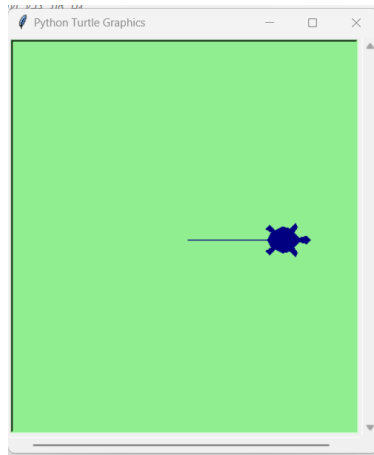
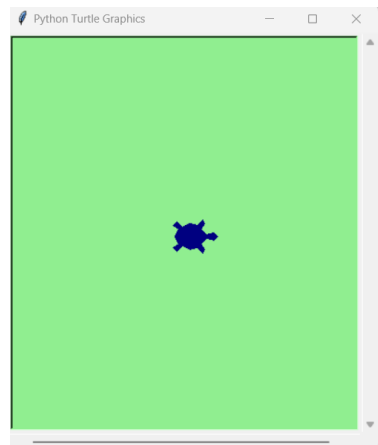
התקדמות קדימה\אחורה

```
# הדמות מתקדמת לכיוון שהיא פונה אליו  
# הפקודה מקבלת כפרמטר את מספר הפיקסלים שמייצגים את המרחק אותו עוברת  
# הדמות מציירת קו תוך כדי ההתקדמות  
# הקו מקבל כברירת מחדל את צבע הדמות. אבל גם את הצבע והעובי של הקו ניתן לשנות בנפרד  
player.forward(100)  
player.backward(50)
```



ריבוע בשלבים

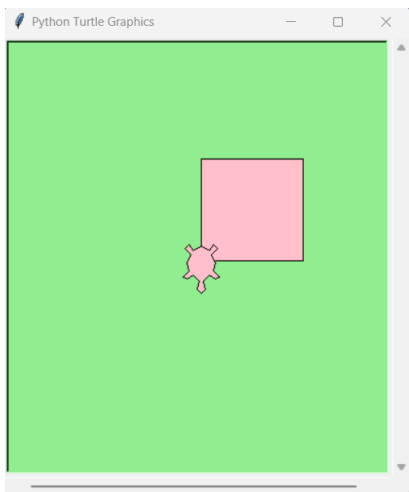
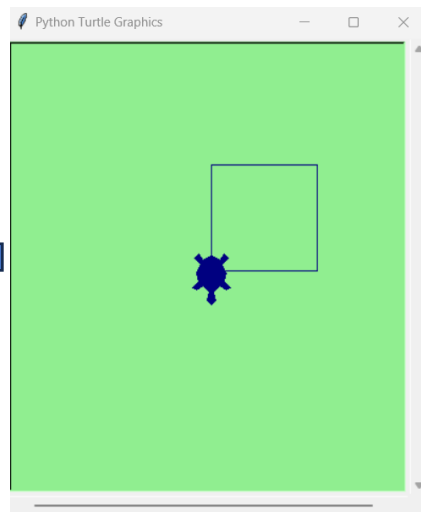
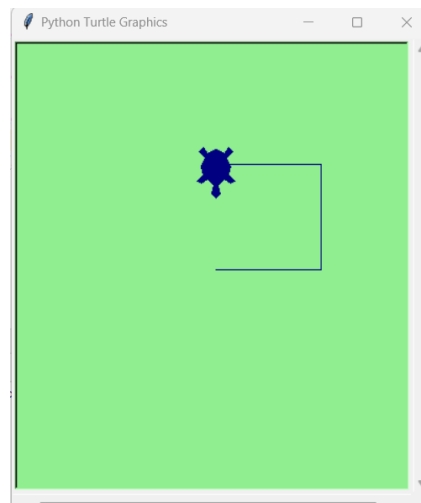
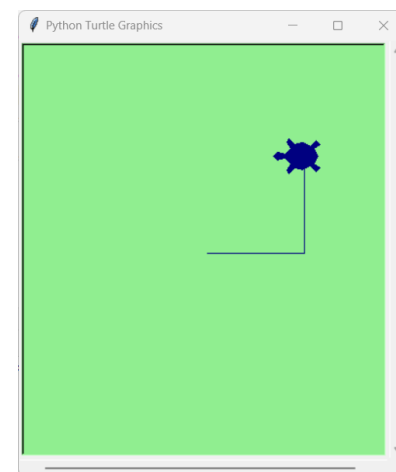
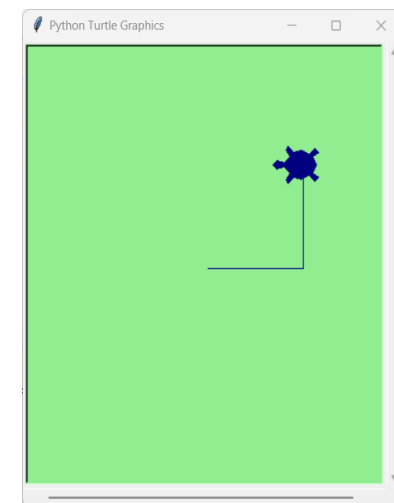
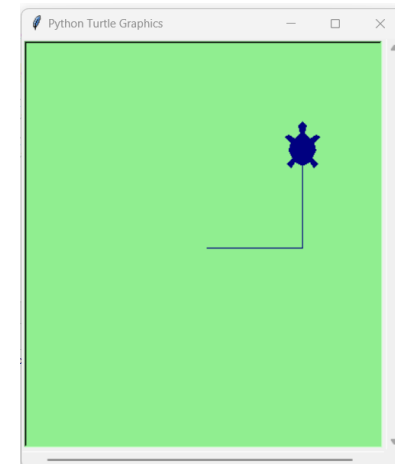
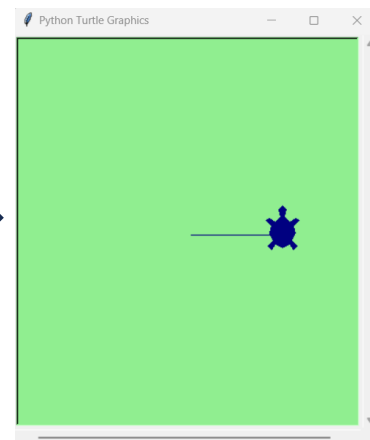
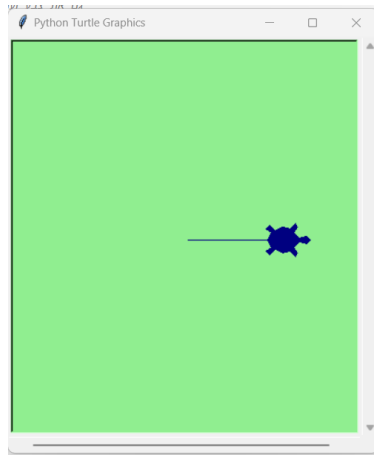
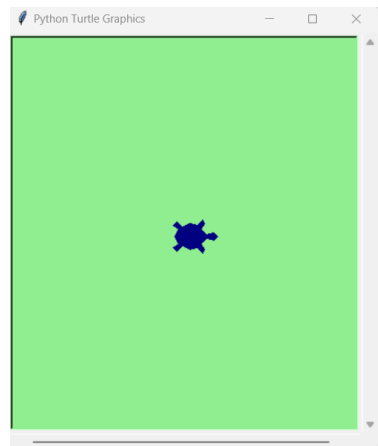
```
player.forward(100)  
player.left(90)  
player.forward(100)  
player.left(90)  
player.forward(100)  
player.left(90)  
player.forward(100)
```



לצבוע את הצורה הנוצרת

```
player.fillcolor('Pink')  
player.begin_fill()
```

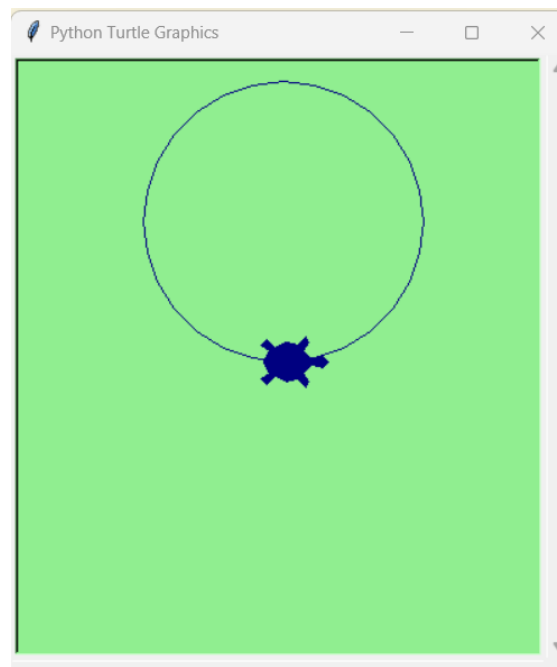
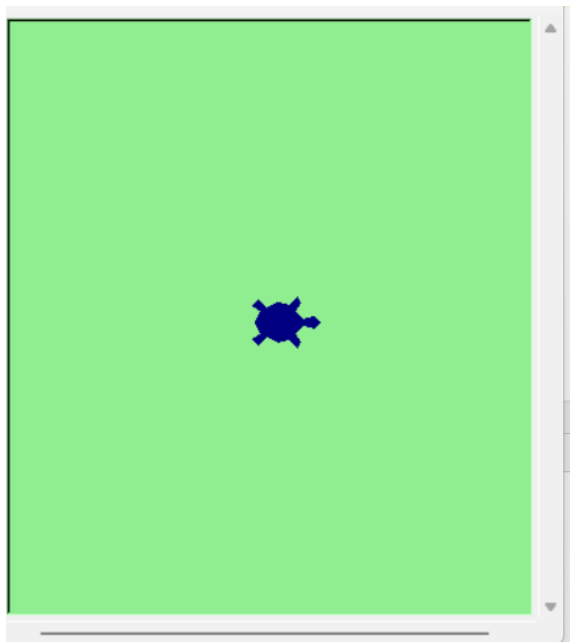
```
player.forward(100)  
player.left(90)  
player.forward(100)  
player.left(90)  
player.forward(100)  
player.left(90)  
player.forward(100)  
player.end_fill()
```



עיגול

הפרמטר הוא רדיוס העיגול

```
player.circle(100)
```

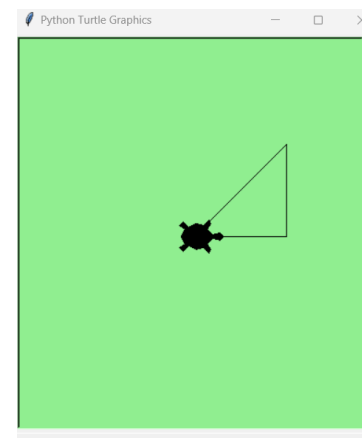
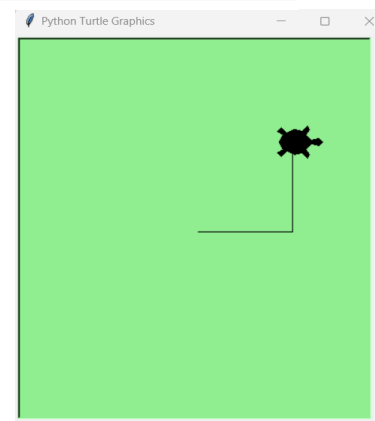
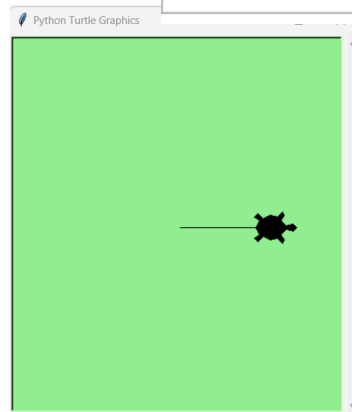
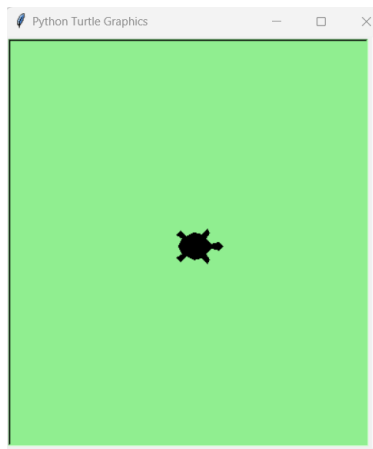


פקודת goto

```
#X ציר  
player.goto(100,0)  
#Y ציר  
player.goto(100,100)  
#לחזור לנקודת התחלה  
player.goto(0,0)
```

X<0 Y>0	X>0 Y>0
X<0 Y<0	X>0 Y<0

ראשית
הצירים



להרים את העט

```
#pen up  
player.pu()  
player.goto(100,0)  
#pen down  
player.pd()
```

